scaled horizontally without limits, because they add more machines to the cluster if additional resources are needed.

10.6.1 Increasing database resources

When you start an RDS database, you choose an instance type. The instance type defines the computing power and memory of your virtual machine (such as when you start an EC2 instance). Choosing a bigger instance type increases computing power and memory for RDS databases.

You started an RDS database with instance type `db.t2.micro`, the smallest available instance type. You can change the instance type using a CloudFormation template, the CLI, the Management Console, or AWS SDKs. You may want to increase the instance type if performance is inadequate. You will learn how to measure performance in section 10.7. Listing 10.8 shows how to change the CloudFormation template to increase the instance type from `db.t2.micro` with one virtual core and 615 MB memory to `db.m3.large` with two faster virtual cores and 7.5 GB memory. You'll do this only in theory. Don't do this to your running database because it is not covered by the Free Tier and will incur charges. Keep in mind that modifying the instance type causes a short downtime.

Listing 10.8 Modifying the instance type to improve performance of an RDS database

```
Database:
  Type: 'AWS::RDS::DBInstance'
  DeletionPolicy: Delete
  Properties:
    AllocatedStorage: 5
    BackupRetentionPeriod: 3
    PreferredBackupWindow: '05:00-06:00'
```